

Face-to-BMI Prediction using Transfer Learning and Multi-Task Deep Learning

Anuradha Rakh

May 24, 2026

Abstract

This project develops an end-to-end machine learning system for predicting Body Mass Index (BMI) from facial images. The work is inspired by the Face-to-BMI paper, which reported an overall Pearson correlation of 0.65 using VGG-Face features and support vector regression. We implemented and compared four deep learning models: a VGG16 baseline, a ResNet50 multi-task model, an EfficientNet fine-tuning model, and a ResNet50 BMI-only ablation. The best model was the ResNet50 multi-task architecture, which jointly predicted BMI and gender, achieving a Pearson correlation of 0.6811 and surpassing the reported benchmark. The project also includes a Streamlit-based user interface for image upload, webcam input, model selection, and BMI prediction display.

1 Introduction

Body Mass Index (BMI) is a widely used measure for estimating body weight status using height and weight. Traditional BMI measurement requires explicit height and weight information, but recent advances in computer vision suggest that facial appearance may contain useful visual signals correlated with BMI. Facial adiposity, jaw structure, cheek fullness, and other visual features may provide predictive information.

The goal of this project is to build a practical deep learning system that predicts BMI from face images. The project follows a research-style workflow: dataset preparation, model development, training, evaluation, hyperparameter tuning, ablation study, and deployment through a web interface.

2 Problem Statement

Given an input facial image, the objective is to predict a continuous BMI value:

$$f(x) \rightarrow \hat{y}_{BMI}$$

where x is the input image and $\hat{y}_{BMI}$ is the predicted BMI. This is a regression problem, not a classification problem.

The main evaluation metric is Pearson correlation coefficient r , which measures how strongly the predicted BMI values correlate with the true BMI values.

3 Dataset

The dataset contains facial images and metadata including BMI, gender, train-test split indicator, and image filename.

3.1 Dataset Format

The CSV file contains the following columns:

- **bmi**: continuous BMI label
- **gender**: Male or Female
- **is_training**: train/test split indicator
- **name**: image filename

3.2 Dataset Statistics

Metric	Value
Total CSV rows	4206
Valid usable images	3962
Training samples	3210
Testing samples	752
Male samples	2354
Female samples	1608

Table 1: Final dataset statistics after removing missing images.

4 Overall Project Pipeline

Figure 1 shows the complete project workflow.

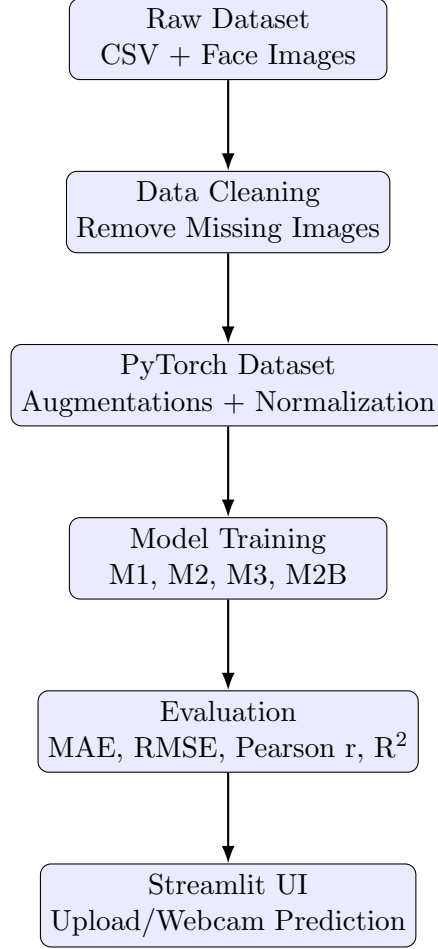


Figure 1: End-to-end Face-to-BMI project pipeline.

5 Methodology

This project uses transfer learning. Instead of training a convolutional neural network from scratch, pretrained models are reused as visual feature extractors. A small task-specific prediction head is added on top of each backbone.

The general model structure is:

$$\text{Image} \rightarrow \text{Pretrained CNN Backbone} \rightarrow \text{Regression Head} \rightarrow \text{BMI}$$

6 Model Architectures

6.1 M1: VGG16 Baseline

The first model uses VGG16 pretrained on ImageNet. The convolutional backbone is frozen, and a dense regression head is trained for BMI prediction.

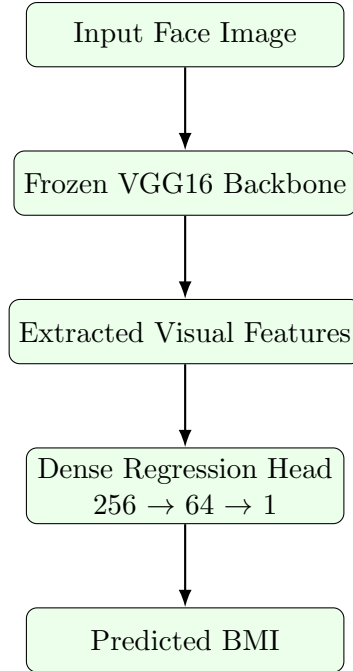


Figure 2: M1 VGG16 baseline architecture.

6.2 M2: ResNet50 Multi-Task Model

The best-performing model uses ResNet50 as a shared backbone. It predicts both BMI and gender. BMI prediction is the primary task, while gender classification is an auxiliary task.

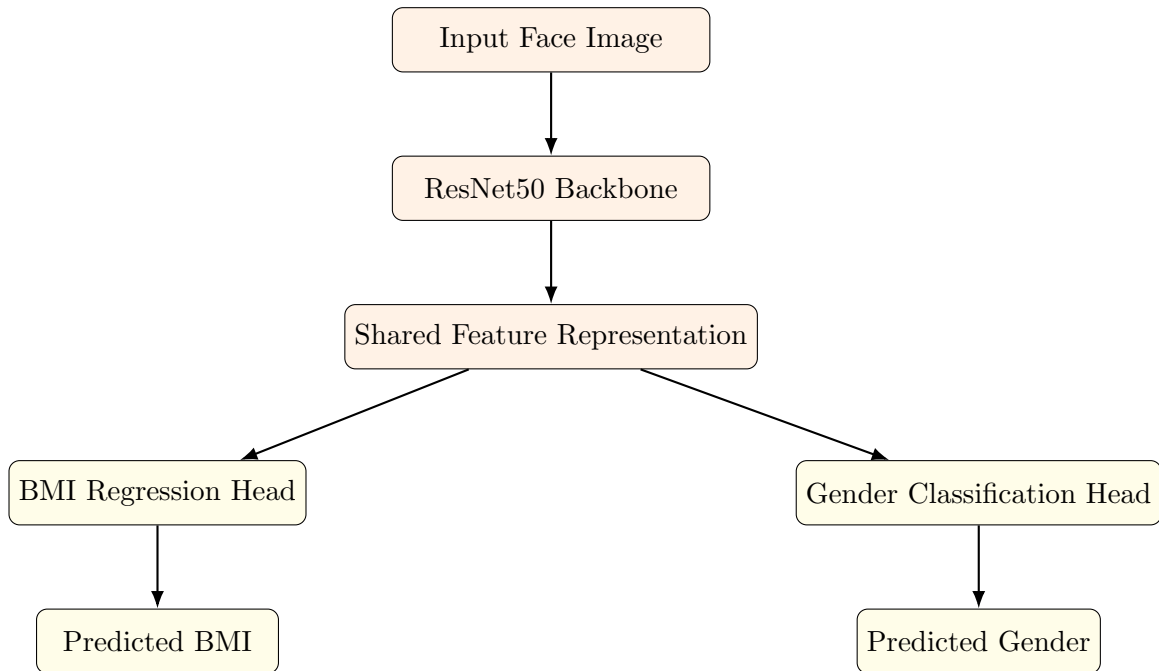


Figure 3: M2 ResNet50 multi-task architecture.

The total loss is:

$$L = \lambda_{BMI} L_{BMI} + \lambda_{gender} L_{gender}$$

where L_{BMI} is the regression loss and L_{gender} is the gender classification loss.

The final tuned configuration used:

- Epochs: 35
- Learning rate: 0.00003
- Dropout: 0.25
- BMI loss weight: 1.0
- Gender loss weight: 0.10

6.3 M3: EfficientNet Fine-Tuning

EfficientNet-B0 was used as a modern CNN backbone. Most layers were initially frozen, while the last three blocks were unfrozen for fine-tuning.

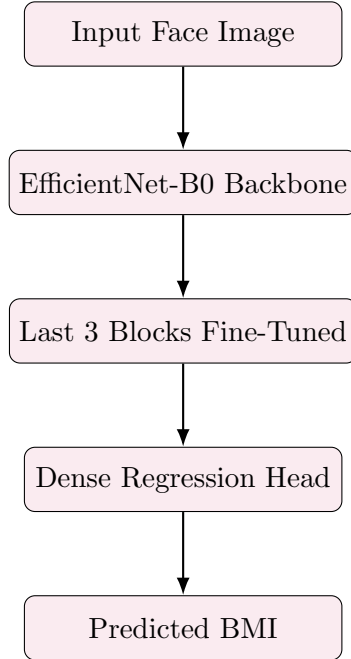


Figure 4: M3 EfficientNet fine-tuning architecture.

6.4 M2B: ResNet50 BMI-Only Ablation

The M2B model was created as an ablation study. It uses the same ResNet50 backbone but removes the gender classification branch. This tests whether auxiliary gender supervision improves BMI prediction.

7 Training Strategy

The training process used the following steps:

1. Load cleaned train/test CSV files.
2. Apply training image augmentations.
3. Normalize images using ImageNet statistics.
4. Train the selected model.
5. Evaluate on the held-out test set.
6. Save the checkpoint with the best Pearson correlation.

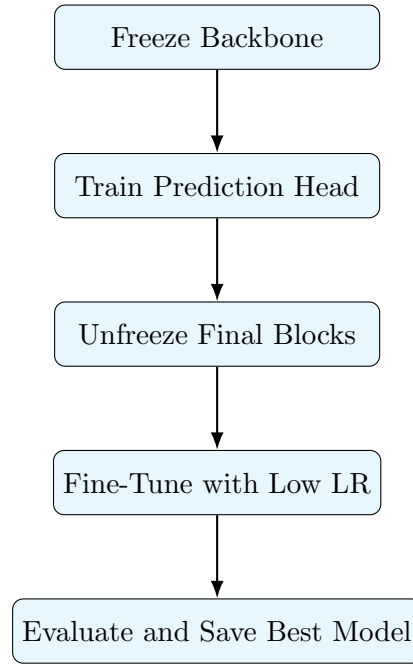


Figure 5: Training and fine-tuning workflow.

8 Evaluation Metrics

The primary metric is Pearson correlation coefficient:

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}$$

Additional metrics include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 score

9 Experimental Results

Model	Description	Pearson r
M1	VGG16 Baseline	0.4227
M3	EfficientNet Fine-Tune	0.5532
M2B	ResNet50 BMI-Only	0.5771
M2	ResNet50 Multi-Task	0.6811

Table 2: Final model comparison.

The ResNet50 multi-task model achieved the best result with a Pearson correlation of 0.6811. This surpassed the paper benchmark of 0.65.

10 Discussion

The results show that ResNet50 features are more effective than VGG16 features for this task. The multi-task approach slightly outperformed the BMI-only ResNet50 ablation, suggesting that gender supervision acted as a useful regularizer.

EfficientNet improved after tuning but did not outperform ResNet50. This may be due to the limited size of the dataset, sensitivity of EfficientNet to hyperparameters, or the need for stronger face-specific preprocessing.

11 Streamlit Deployment

A Streamlit user interface was created for demonstration. The interface supports image upload, webcam input, model selection toggles, gender selection, prediction results, loading spinner, and reset buttons.

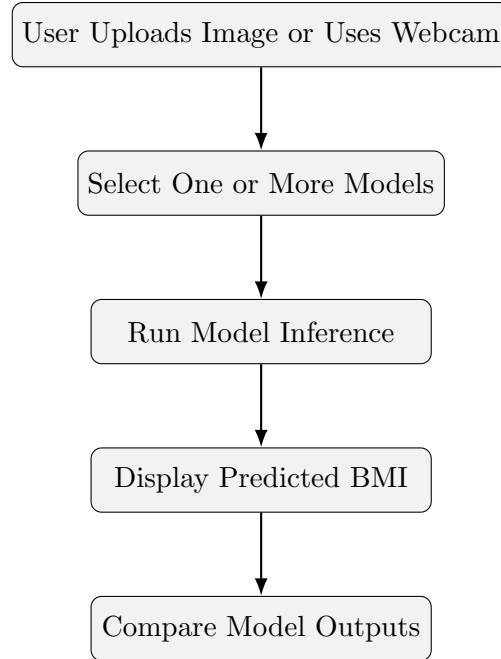


Figure 6: Streamlit deployment workflow.

12 Limitations

Although the final model surpassed the reported benchmark, several limitations remain:

- BMI from face alone is noisy and imperfect.
- Lighting, pose, age, expression, and image quality affect predictions.
- Dataset size is relatively small for deep learning.
- BMI is a population-level measure and should not be treated as a complete individual health assessment.
- Ethical concerns exist when inferring body-related attributes from images.

13 Future Work

Future improvements could include:

- Face alignment before model input
- ArcFace or FaceNet embeddings
- Ensemble averaging
- Better hyperparameter search
- Grad-CAM visualization
- Model calibration
- More diverse training data

14 Conclusion

This project built a complete Face-to-BMI prediction system using deep learning. Four models were implemented and compared. The best model, ResNet50 multi-task learning, achieved a Pearson correlation of 0.6811, outperforming the benchmark value of 0.65. The project demonstrates transfer learning, fine-tuning, multi-task learning, model comparison, and deployment through a Streamlit interface.

References

Kocabey, E., Camurcu, M., Ofli, F., Aytar, Y., Marin, J., Torralba, A., and Weber, I. Face-to-BMI: Using Computer Vision to Infer Body Mass Index on Social Media. ICWSM, 2017.